Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **React/TypeScript products**, **Python APIs**, real-time consumer workflows, and **LLM-powered experiences** supported by scalable data and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant LLM services** with explicit state transitions, failure isolation, recoverable processing, and observable workflows under sustained production load.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Engineered an **AI-powered grading platform** using GPT-4o, Claude, Qwen-VL, and OCR to evaluate 10K+ submissions with 90%+ scoring consistency across text and handwritten assessment workflows at production scale.
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with WebSockets and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **automated testing and deployment**, implementing JWT access controls, WebSocket updates, Azure monitoring, Docker/Jenkins pipelines, and ARM templates across eight modular service domains.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web booking workflows while reducing design errors by 25% through reusable components.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, C++, Bash
Frontend & Mobile: React, React Native, Next.js, HTML, CSS, Responsive Design
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets, Microservices
AI & Product: OpenAI API, Claude, GPT-4o, Qwen-VL, LLM Systems, Transformers, Model Evaluation
Data & Cloud: PostgreSQL, MySQL, MongoDB, Azure SQL, AWS RDS, Firebase, Query Optimization
Delivery & Quality: Docker, Kubernetes, Jenkins, CI/CD, Git, GitHub, Unit Testing, Integration Testing, Automated Testing, PyTest, Debugging, Observability, Agile

Projects

AI Event Discovery Experience - Design Concept
- Designed an **AI event-discovery concept** using React, TypeScript, FastAPI, LLM intent parsing, live inventory APIs, preference refinement, localization, and explainable recommendations for conversational search across fan journeys.

Consumer Experimentation Platform - Design Concept
- Designed a **consumer experimentation platform** concept with feature variants, event tracking, conversion funnels, localized interfaces, checkout states, performance monitoring, and reproducible analysis supporting iterative product decisions reliably.

FinNexus | Live
- Built a **secure TypeScript fintech platform** supporting 50K+ users and 1M+ transactions through Plaid integration, token-based access controls, real-time synchronization, Kubernetes deployment, and MongoDB-backed filtering workflows reliably.